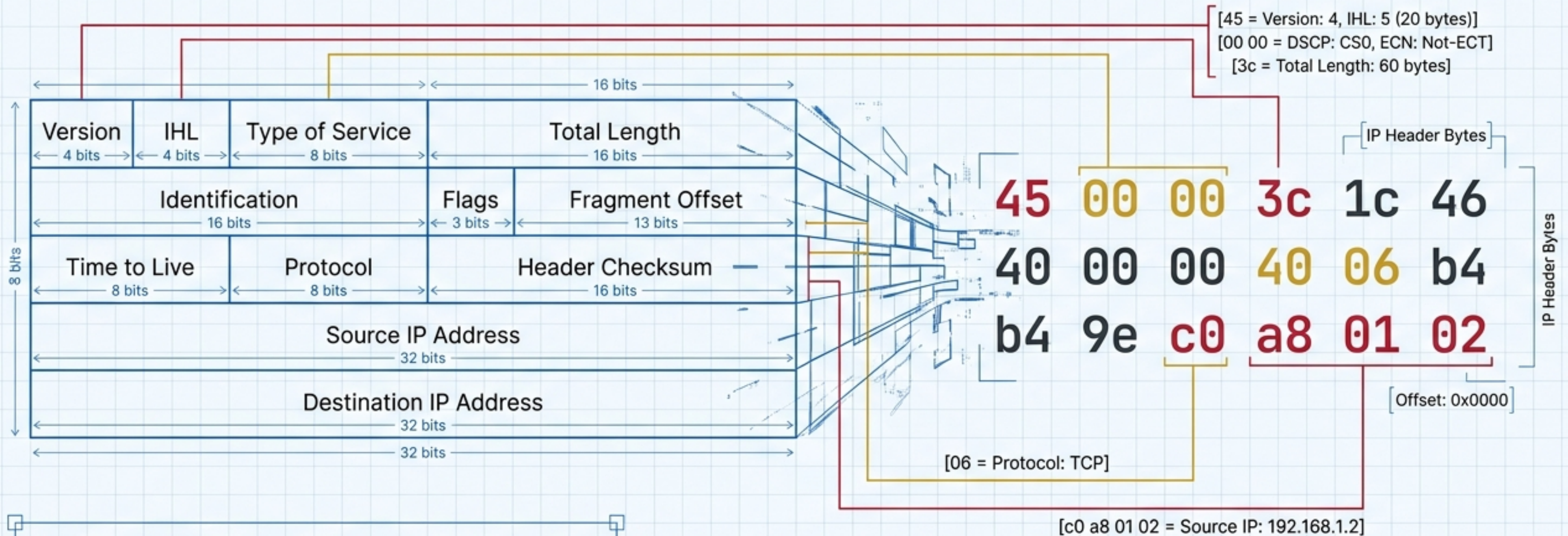


From RFCs to Raw Bytes

Deconstructing Network Protocols



A Visual Guide to Standards, Headers,
and Byte-Level Packet Parsing.

The Three Pillars of a Protocol

GET / HTTP/1.1

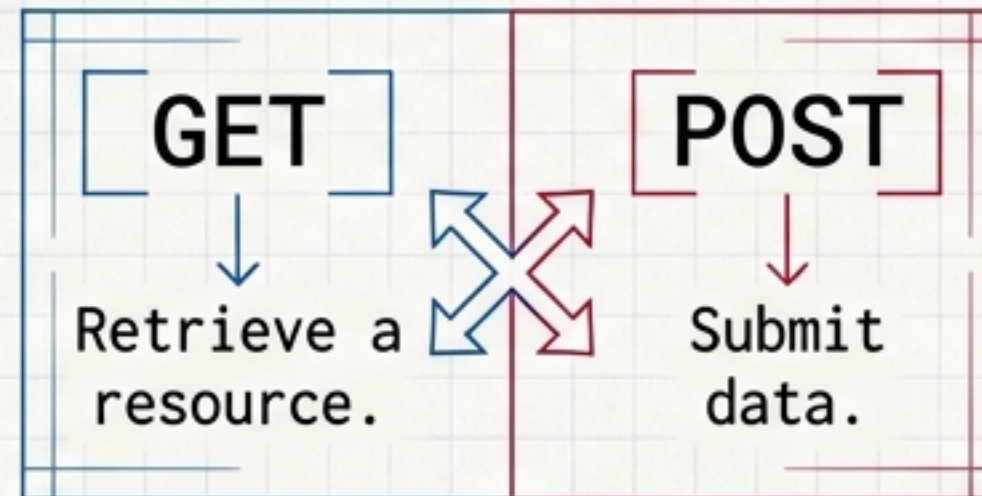
Syntax (Format)



The fixed, non-negotiable order of data.

- Method (GET) -> Path (/) -> Version (HTTP/1.1).

Semantics (Meaning)



What each specific field instructs the receiver to do.

- GET: Retrieve a resource.
- POST: Submit data.

Same syntax position, entirely different meaning.

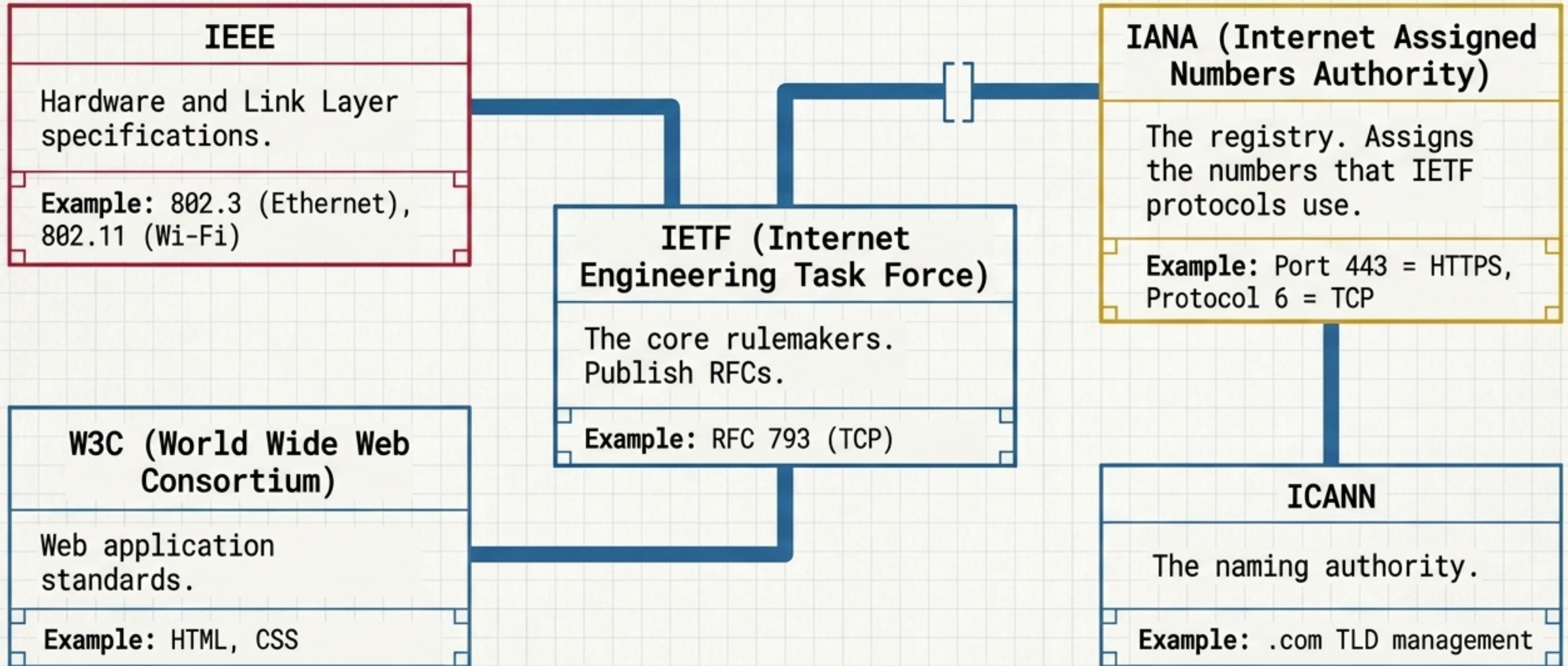
Timing (Sequence)



The mandatory order of messages.

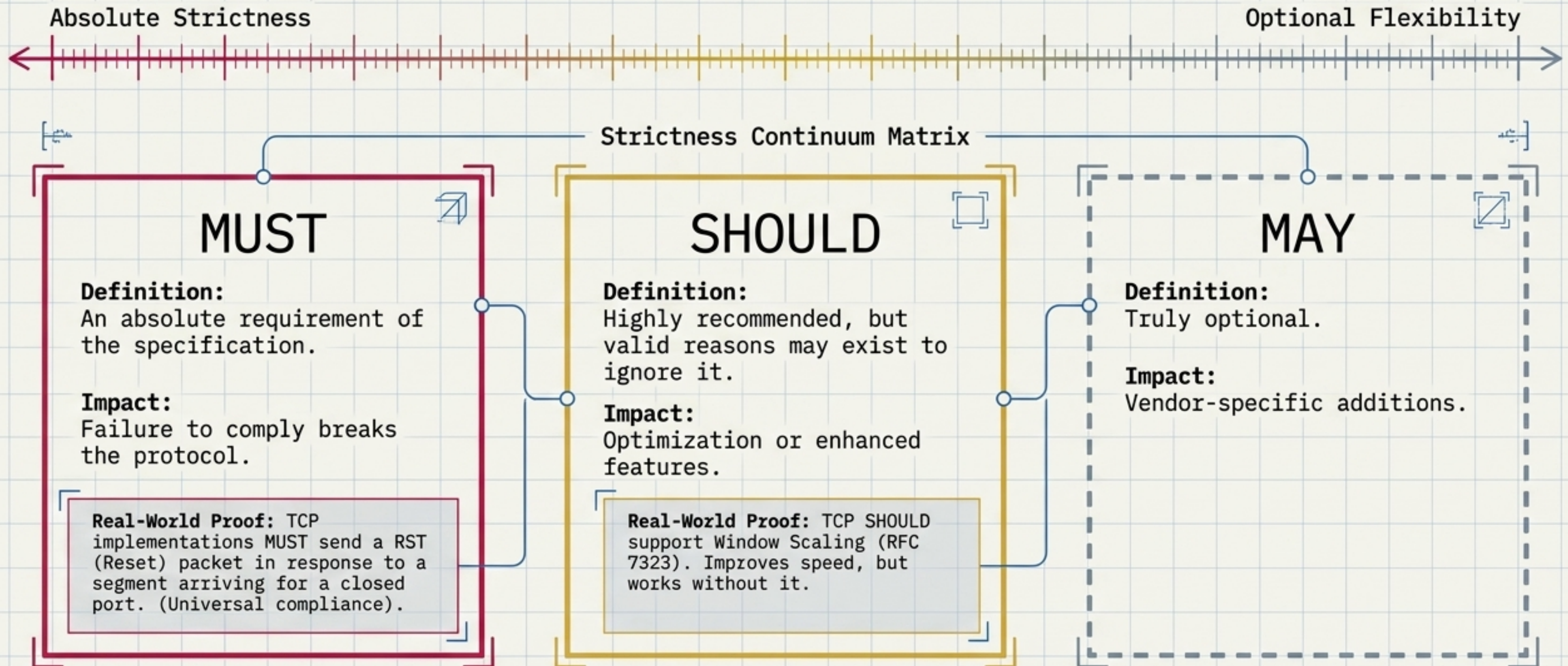
- A TCP connection MUST be established before any HTTP data is transmitted.

The Internet Rulemakers: Governance Ecosystem



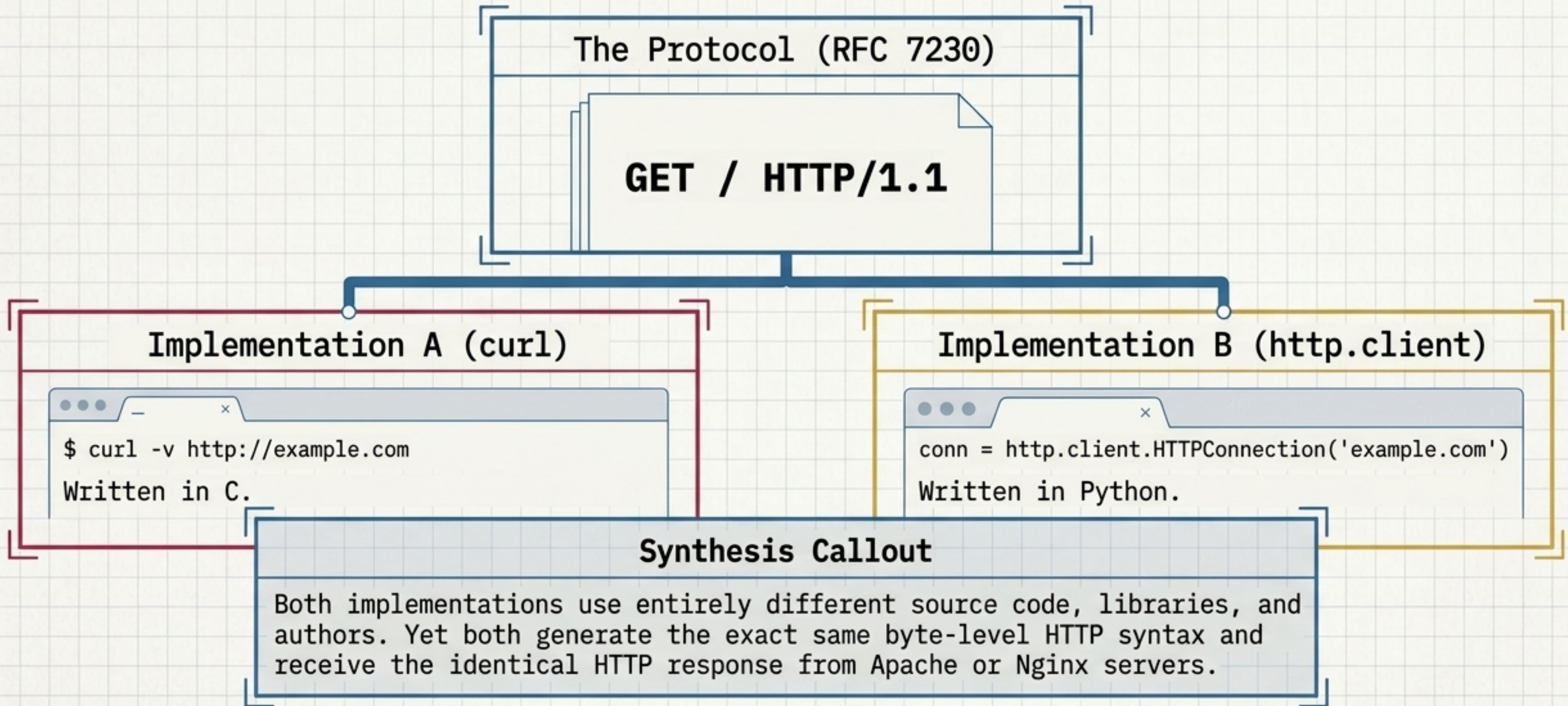
The Language of Interoperability (RFC 2119)

How to read the keywords that govern network behavior.



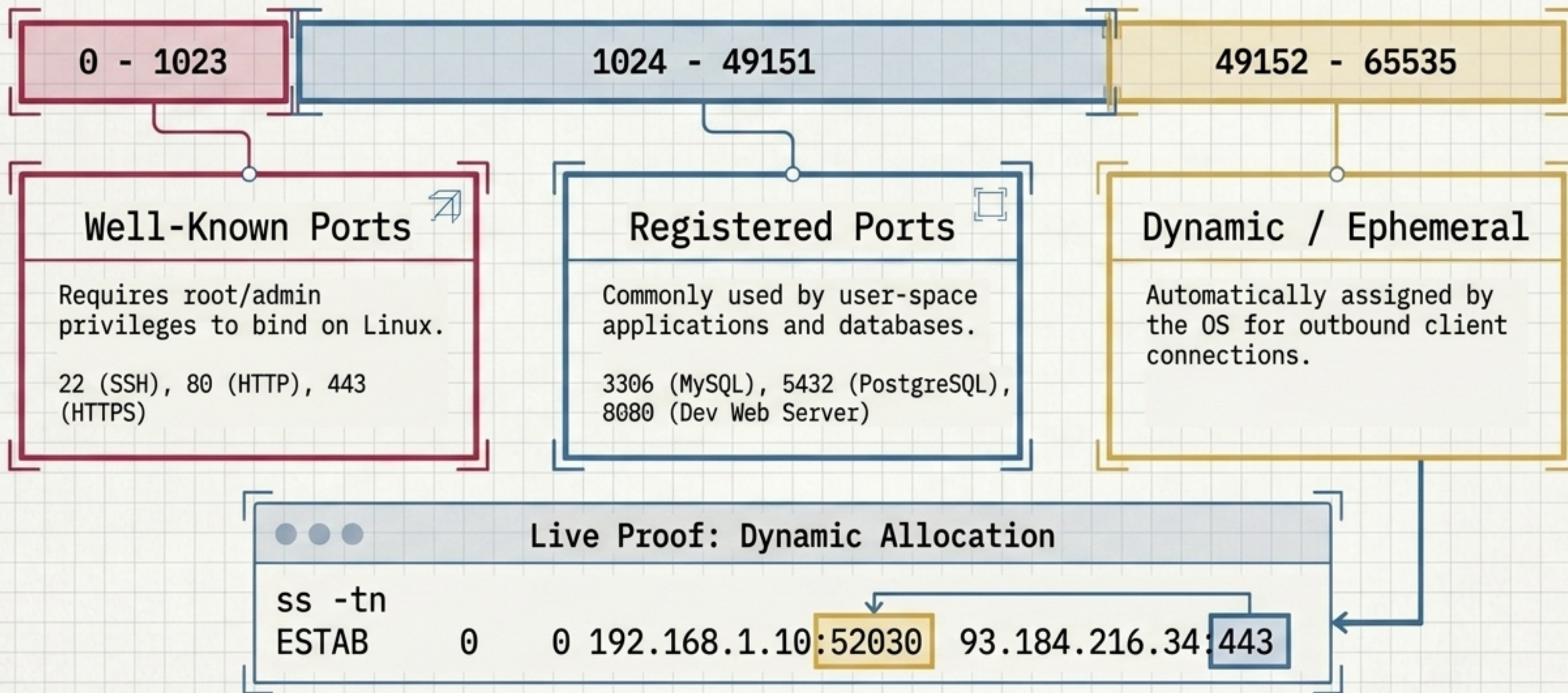
Protocol ≠ Implementation

The protocol is the rulebook. The implementation is the code.



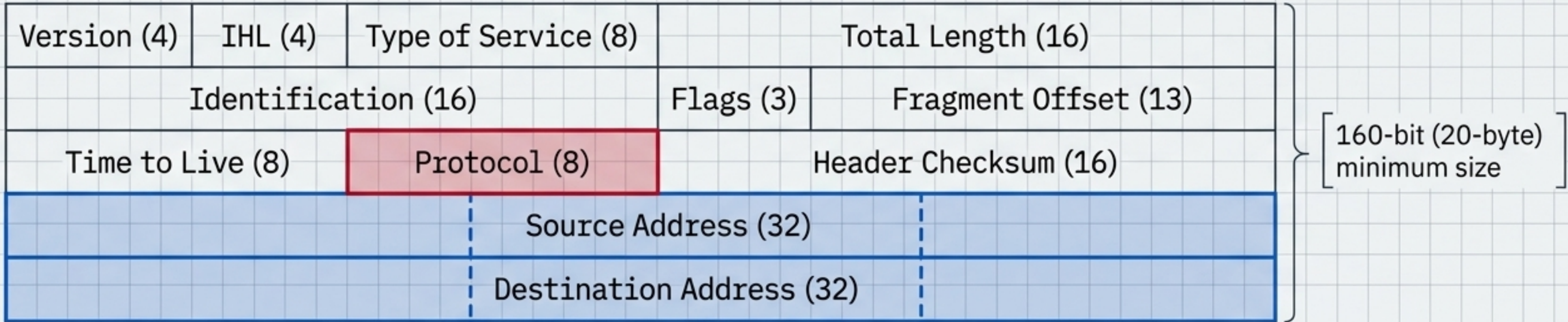
The IANA Port Taxonomy

Classifying the 65,535 available network addresses.



Deconstructing IPv4 (RFC 791)

[The Blueprint]



[The Reality]

Internet Protocol Version 4, Src: 192.168.1.10, Dst: 93.184.216.34

...

Time to Live: 64

Protocol: TCP (6)

Header Checksum: 0xb49e

[Insight Callout]

The Protocol field is IANA's registry in action. Without centralized assignment, vendors would use different numbers to represent TCP, instantly breaking IP communication.

Deconstructing TCP (RFC 793)

The binary switches that control connection state.

TCP Header			Minimum size: 20 bytes
Source Port (16)			
Destination Port (16)			
Sequence Number (32)			
Acknowledgment Number (32)			
Data Offset (4)	Reserved (6)	Flags (Control Bits) (6)	
Window (16)			
Checksum (16)			
Urgent Pointer (16)			
Options (Variable)		Padding	

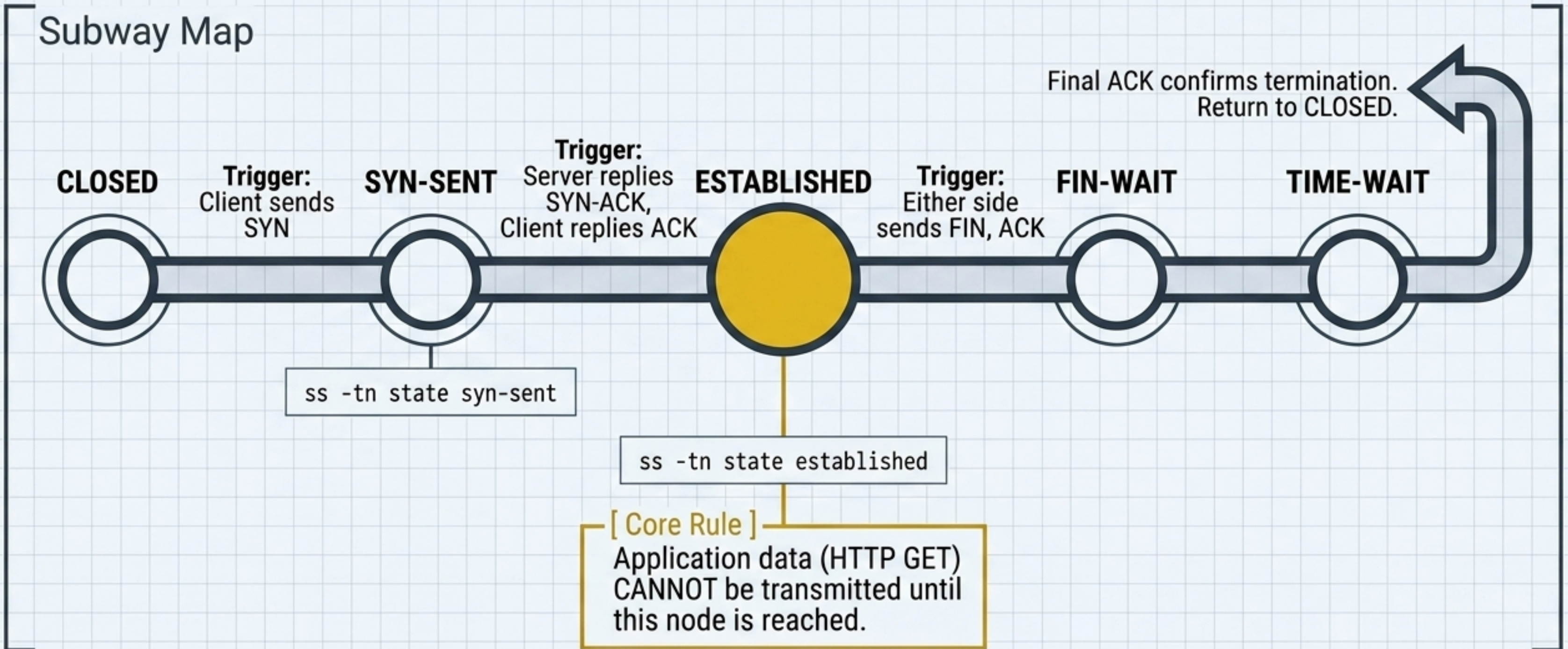


The Control Bits Matrix			
Flag (Meaning)	SYN Packet	SYN-ACK Packet	
URG (Urgent data)	0	0	
ACK (Acknowledgment field valid)	0	1	
PSH (Push data to app)	0	0	
RST (Reset connection)	0	0	
SYN (Synchronize/Initiate)	1	1	
FIN (Finish/No more data)	0	0	

Key Takeaway: A TCP Handshake is not magic; it is simply the sequential transmission of packets with these specific bits physically flipped to 1.

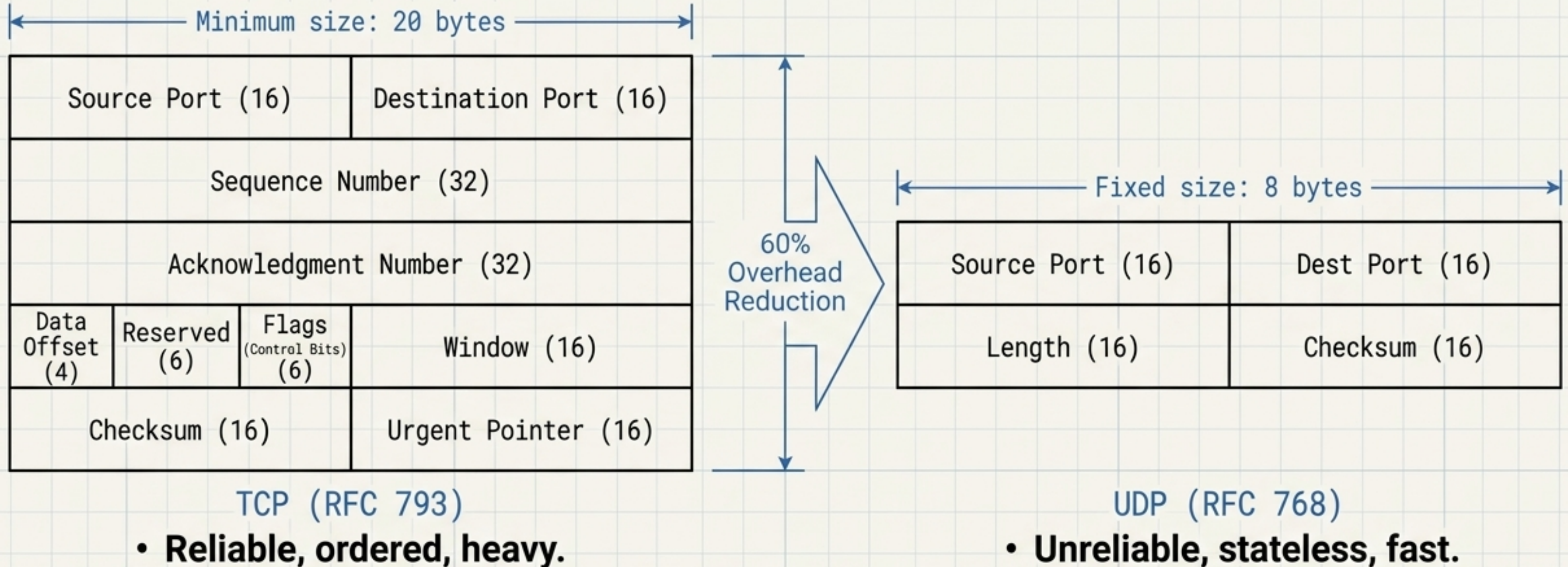
The TCP State Machine in Action

Tracking connection lifecycle from initialization to termination.



TCP vs. UDP: The Cost of Reliability

Why DNS relies on intentional simplicity.



UDP is 60% smaller than TCP
(saves 12 bytes per datagram).

At 1 billion DNS queries a day, UDP's intentional simplicity
saves 12 Gigabytes of bandwidth overhead every 24 hours.

Anatomy of an ARP Request (RFC 826)

Resolving IP addresses to physical MAC hardware.

Layer 1: Ethernet Header (14 Bytes)

Destination MAC: ff:ff:ff:ff:ff:ff (Annotation: Broadcast to all switch devices)

Source MAC: Sender's known MAC

EtherType: 0x0806 (Annotation: Parse payload as ARP)

Layer 2: ARP Payload (28 Bytes)

Hardware: Ethernet (1)

Protocol: IPv4 (0x0800)

Operation: 1 (Request)

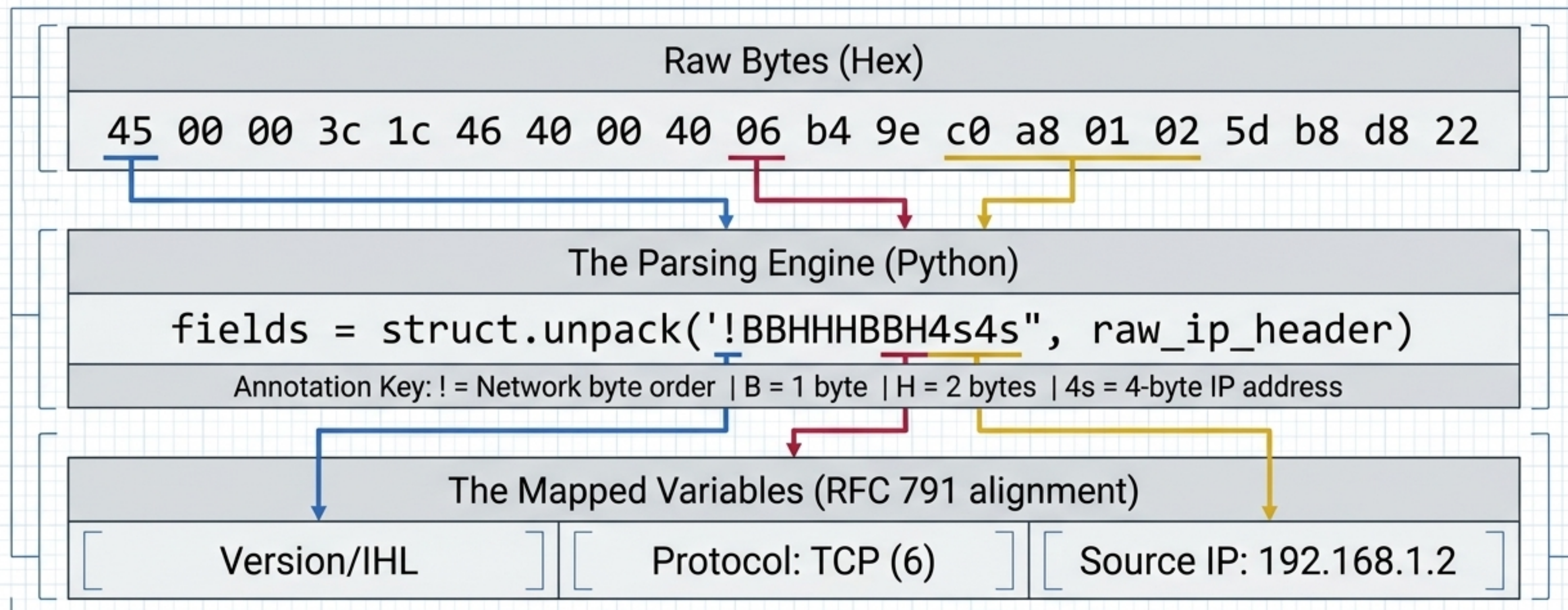
The Knowns: Sender MAC, Sender IP, Target IP.

The Unknown: Target MAC = 00:00:00:00:00:00

This string of zeros represents the missing variable the protocol is broadcasting to solve.

The Kernel-Level View

How operating systems translate electrical signals into data structures.



This single line of code is the literal translation of the RFC 791 block diagram into working software.

Foundational Protocol Cheat Sheet

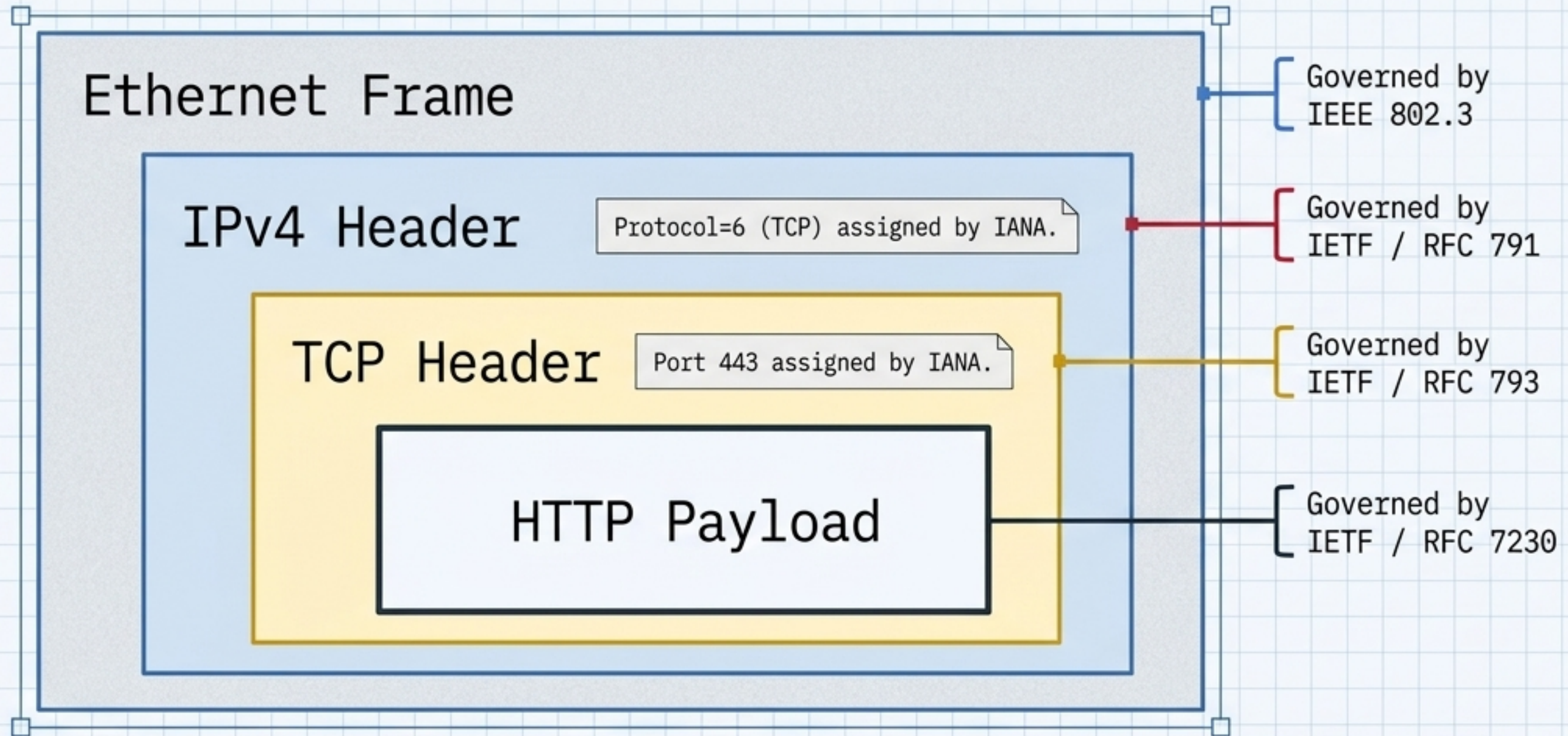
Core internet specifications and their authorities.

Protocol	Body	Document	Purpose
IPv4	IETF	RFC 791	Defines IP addressing and packet routing.
TCP	IETF	RFC 793	Reliable, connection-oriented data delivery.
UDP	IETF	RFC 768	Stateless, low-overhead datagram transmission.
ARP	IETF	RFC 826	Resolves network layer IPs to link layer MACs.
DNS	IETF	RFC 1035	Domain name to IP address resolution.
TLS 1.3	IETF	RFC 8446	Cryptographic protocol for secure communication.
Ethernet	IEEE	802.3	Hardware specification for wired LANs.

Note: RFC 2119 governs the **MUST/SHOULD/MAY** keyword compliance across all IETF documents.

Synthesis: The Nested Network

A single web request utilizes multiple protocols operating in perfect sequence.



Core Insight: To successfully render a single webpage, hardware protocols (IEEE) must carry routing packets (IETF), targeting specific application ports (IANA), delivering syntax-perfect payloads (W3C/IETF).

The Cybersecurity Imperative

You cannot defend, exploit, or analyze a network until you can read its foundational blueprints.

1. Tools Lie; Packets Don't.

Abstractions and GUIs hide critical diagnostic data. The truth of a network intrusion or failure always exists at the byte level.

2. Implementation is the Vulnerability.

Protocols are just rules. Exploits target how a developer actually wrote the C or Python code to parse those rules.

3. Mastery Over Memorization.

Understanding *why* a TCP header is 20 bytes, or *how* a bit-flip changes a connection state, separates tool-users from system engineers.